Vincent Do

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Experience

Boeing Company — Berkeley, MO
Software Engineer — *Aug 2024 – Present*

- Developed production C++ features delivering real-time message streams to embedded display systems, with automated tests and Dockerized build workflows for deployment
- Owned 3+ integration and release readiness for a shared framework consumed by 60+ dependent modules, resolving compatibility issues to maintain timing accuracy in real-time message delivery
- Identified root cause of long-standing cross-system defect within two weeks of onboarding to unfamiliar systems, tracing a binary data merge edge case and restoring correct downstream output behavior
- Spearheaded month-long evaluation of Lockheed Martin software through custom integration with messaging architecture, verifying customer requirements at 1.5x the required message rate and documenting procedures for future releases

Expedia Group — Chicago, IL
Software Development Engineer Intern — *May 2023 – Aug 2023*

- Built customer-facing Expedia mobile search features, including VIP Access badges, sort/filter options, and improved error handling to reduce booking search friction
- Implemented GraphQL schema changes across multiple Kotlin microservices, enabling new mobile booking flows across backend and client surfaces
- Built backend support for localized search components across 16 languages, partnering with frontend and localization teams to validate dynamic rendering before release

University of Wisconsin–Madison — Madison, WI
Biomedical Undergraduate Researcher — *May 2021 – Aug 2021*

- Analyzed 16,813 genes across 169 phenotypes using Python/R, graph traversal, and visualization to identify phenotype-gene associations
- Reduced search space to 25 high-priority Zellweger candidate genes using frequency thresholds, chromosomal-location filtering, and recursive phenotype hierarchy traversal

Projects

Playable Chess AI | *Python, PyTorch, CUDA, Flask, Pygame, Stockfish* — playable-chess-AI

- Trained CNN+LSTM legal move-policy model on 4.18M GM/Magnus positions, achieving 71.2% top-5 accuracy on a 390K-position held-out test set
- Engineered GPU-accelerated PyTorch training pipeline with mixed precision and a custom IterableDataset, reducing per-epoch training time from 20+ hours on CPU to 15–25 minutes
- Engineered PGN preprocessing pipeline for zipped datasets, deduplicating games, computing centipawn-loss metadata, generating legal-move masks, and producing compressed train/validation/test chunks

Shortest Path Between Airports | *C++, React* — airports-global

- Implemented BFS, Floyd–Warshall, and betweenness centrality algorithms to analyze connectivity and routing patterns in aviation networks, visualizing top 10 most important airports using WebGL-based globe rendering

Exploding Chickens | *Node.js, MongoDB* — chickens.rakerman.com

- Built a real-time multiplayer backend with asynchronous game logic and persistent session state, ensuring consistent gameplay across concurrent players with 1000+ games played

Education

University of Illinois Urbana-Champaign — Champaign, IL
B.S. Computer Science & Chemistry — GPA: 3.6 — *Aug 2020 – May 2024*

Technical Skills

Languages: C++, Python, Java, C#, JavaScript, TypeScript, SQL, Kotlin
Frameworks/Tools: React, Node.js, Flask, PyTorch, CUDA, REST APIs, GraphQL, Docker, Jenkins, Git, Linux, MongoDB, PostgreSQL, GCP